

VIVAN DESAI

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EDUCATION

University of Toronto

Bachelor of Science, Specialist in Computer Science

Toronto, Ontario

Class of 2028

PROJECTS

Finsnap | Python, FastAPI, PostgreSQL, React, scikit-learn, JWT | finsnap-six.vercel.app

May 2026

- Built a full-stack transaction classification app using a scikit-learn MLP trained on 30,000 transactions across 3,333 merchants and 23 categories, achieving 88% weighted F1 accuracy across Canadian and GCC merchants.
- Productionized the ML pipeline as a REST API using FastAPI with JWT authentication and PostgreSQL; deployed backend on Railway with a React frontend on Vercel.
- Implemented a user feedback loop where misclassification corrections are logged to a corrections table and used to retrain the model, same pattern used in production ML systems.

Predicting Customer Churn Using Random Forest & Hyperparameter Optimization

Aug 2023 – Mar 2024

- Built a full supervised ML workflow to predict customer churn in a telecom service; cleaned and engineered features from a 7,000-row dataset, applied Random Forest classification with RandomizedSearchCV hyperparameter tuning across five parameters.
- Improved cross-validation score from 78.9% to 81.0% with gains across accuracy, precision, recall, and F1. Awarded highest band (A) for the IB Extended Essay.

PROFESSIONAL EXPERIENCE

Bank Muscat

Business Intelligence Intern

Muscat, Oman

Summer 2025, Summer 2026

- Built an end-to-end NLP and ML pipeline to classify retail banking transactions into spending categories, forming the foundation for a Personal Finance Management tool later extended into a deployed full-stack product (Finsnap).
- Conducted comparative benchmarking across MLP, Gradient Boosting, and SVM architectures, optimizing feature selection and scaling pipelines to ensure robust generalization across validation, test, and out-of-sample datasets.
- Presented key insights to the Head of Retail Marketing, translating complex model outputs into actionable cost-efficiency recommendations, resulting in a proposed initiative for automated customer segmentation and transaction-level reporting.

IMTAC Oman

Data Analytics Intern

Muscat, Oman

Summer 2024

- Developed supervised ML models for housing-price prediction using structured economic and spatial features; optimized hyperparameters and feature importance analysis to reduce forecasting error and provide more reliable price insights.
- Collaborated with senior mentors to convert academic ML concepts into a practical pricing prototype, integrating domain heuristics into the feature pipeline. The tool became a reusable use case for future internal prediction projects.

EXTRACURRICULAR EXPERIENCE

Toronto Student Investment Council (TSIC)

Senior Analyst

Toronto, Ontario

Apr 2025 – Present

- Conducted equity research within the TMT sector, building industry theses and contributing key research for weekly discussions.
- Pitched DoubleVerify at the Fall 2025 Buy-Side Night competition, presenting an AI-driven ad-verification investment thesis.
- Collaborated with the team to refine valuation materials, integrating earnings commentary and improving pitching skills.

SKILLS, ACTIVITIES, AND INTERESTS

Skills: Python, Java, JavaScript, React, FastAPI, PostgreSQL, SQLAlchemy, REST APIs, JWT Authentication, Pandas, Scikit-Learn, HTML, CSS, GitHub, LaTeX, Microsoft Office Suite

Activities: Ex-Amateur Footballer (AIFF-registered; Hero Sub Junior I-League semi-finalists; Karnataka State Team), Hiking/Cliff Jumping

Interests: Astrophysics & Physics (Brian Cox podcasts; Richard Feynman lectures), Camping, DJing & electronic/trance music production, Football (Manchester City fan), Global Cultures (lived in 4 countries), Travel (visited 14+ countries)